

Towards data-constrained defect inspection in advanced manufacturing: A lightweight diffusion model adapter for high-fidelity data generation

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ABSTRACT

Deep learning-based vision detection has found wide application in manufacturing quality control processes. However, data scarcity remains a critical factor that limits the effectiveness of these methods. While numerous studies have explored utilizing image generation techniques for expanding the datasets, existing methods struggle to generate effective samples. In this paper, we propose Real-Adapter, a novel framework that integrates the diffusion-based large generative foundation model into industrial data augmentation. The framework generates high-fidelity industrial data with fine-grained defects, thereby substantially improving the efficacy of augmentation for downstream defect detection tasks. Our approach leverages the Stable Diffusion Model by capitalizing on its generative capabilities and incorporates two lightweight adapters called Sense Information-Aware (SIA) and Local and Location Information-Aware (LLIA) branches to extract domain-specific industrial product features and precisely align these control signals with the foundation model. This design addresses two key challenges in applying diffusion-based models to industrial image generation: manufacturing domain knowledge gaps, and insufficient fine-grained detail generation. Furthermore, we introduce a Task-Aware (TA) loss to enhance compatibility with downstream detection. Comparative experiments demonstrate that Real-Adapter outperforms all representative generative models, achieving 45.122 and 42.597 Fréchet Inception Distance (FID) on Impeller Defect (ID) and Cylinder Block Defect (CBD) datasets, which surpass those of the second-ranked model by 16.834 and 14.645, respectively. The generalization capability of Real-Adapter is further validated on the NEU-SEG and Magnetic Tile (MT) datasets. Augmentation experiments confirm its effectiveness in enhancing downstream detection tasks across all four datasets.

1. Introduction

In the modern manufacturing quality inspection and control field, deep-learning-based detection technologies (Chen et al., 2025; Hu et al., 2025) have become indispensable, supporting tasks such as defect detection and intelligent assembly inspection. The success of these methods, however, fundamentally depends on the availability of large-scale, high-quality training data, which enables models to learn discriminative feature representations of target defects. Yet, in real-world industrial production, defective samples are inherently scarce and highly imbalanced,

making it difficult to collect adequate data for training. This scarcity limits the ability of models to generalize across diverse scenarios, thereby reducing detection robustness and accuracy.

To mitigate data insufficiency in industrial production quality inspection, numerous data augmentation methods have been proposed to improve the performance of downstream detection models, including both classical (Shorten & Khoshgoftaar, 2019) and generative approaches (Li et al., 2022; Yang et al., 2021). Classical augmentation techniques, such as scaling, rotation, and flipping, expand the dataset by applying linear transformations to the original images. However, these

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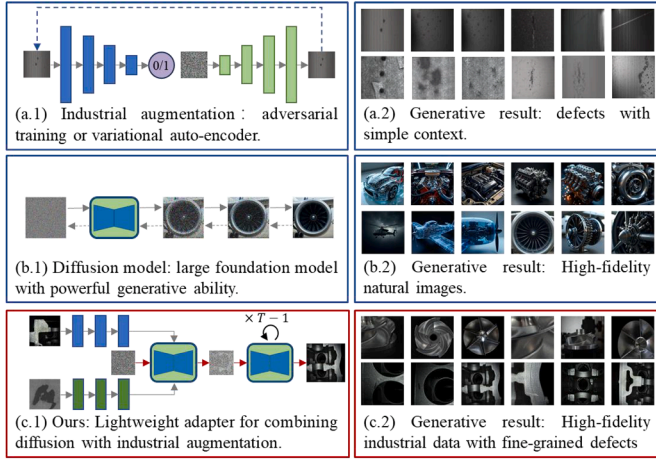


Fig. 1. The motivation for Real-Adapter is to design lightweight control adapters that effectively integrate the strong generative capacity of diffusion-based foundation models, enabling the construction of high-fidelity industrial datasets.

operations only introduce limited variability at the pixel level, without creating genuinely new feature distributions. As a result, they often fail to capture the fine-grained geometric structures and modality diversity that are intrinsic to real industrial data, thereby leading to overfitting and reduced generalization capacity.

Given these limitations, researchers have begun to explore generative models (Huang et al., 2023; Karras et al., 2021; Wang et al., 2018) for synthesizing high-quality industrial image data. For instance, Li et al. (2023) propose a Generative Adversarial Network (GAN) based framework for generating defects with the characteristic of location sensitivity. Ren et al. (2024) propose an unsupervised GAN-based method to address the shortage of paired training data in tunnel defect identification by introducing cycle perceptual consistency loss and feature attention mechanisms. Although these approaches alleviate the challenge of industrial data scarcity to some extent, they often fail to synthesize high-fidelity images of complex industrial parts with intricate backgrounds and diverse defect morphologies, limiting their practical utility in manufacturing settings, as illustrated in Fig. 1(a.2). As a result, a more robust generative paradigm is essential to synthesize image data that faithfully replicates such complexity.

In recent years, many large generative foundation models, such as Imagen (Saharia et al., 2022) and Stable Diffusion (Rombach et al., 2022) grounded in denoising diffusion probabilistic theory (Ho et al., 2020) have garnered significant attention for their outstanding performance in natural image synthesis tasks, such as Artificial Intelligence (AI) painting and artistic creation, image style transfer, etc. Thus, leveraging large diffusion-based generative models presents a promising solution to overcome the limitations inherent in conventional data augmentation paradigms. However, applying the existing diffusion-based generation paradigm (Ho et al., 2020; Ye et al., 2023; Zhang et al., 2023) to industrial data enhancement remains challenging. (1) Existing large pre-trained generative models, primarily trained on natural images, lack the domain-specific knowledge required for industrial applications. Moreover, their vast parameter scales make retraining or fine-tuning with limited industrial data particularly challenging, often leading to sub-optimal adaptation. Consequently, directly applying these models to industrial data generation results in outputs that diverge significantly from real-world scenarios, as illustrated in Fig. 1(b.2). (2) Although diffusion-based models can effectively capture the global structure of images, they often struggle to reproduce fine-grained local details (Aithal et al., 2024). Since industrial data enhancement requires faithful preservation of authentic defect characteristics, this limitation critically undermines their applicability.

To bridge these gaps, we propose Real-Adapter, the lightweight diffusion model adapters for controlling the large generative foundation model to generate high-fidelity industrial data. Instead of full-model fine-tuning, we freeze the foundation model and introduce two lightweight adapters for controlling the generation process. Specifically, the Sense Information-Aware (SIA) branch is proposed to capture global domain-specific industrial scenarios information, while the Local and Location Information-Aware (LLIA) branch is designed to extract fine-grained defect geometries and precise spatial localizations information. These control signals are injected into the U-Net foundation via a decoupled cross-attention mechanism, enabling high-fidelity and controllable industrial data synthesis. Furthermore, we design a Task-Aware (TA) loss to explicitly align the generative process with the performance of downstream detection tasks.

This study makes the following contributions.

- 1) We propose an innovative and general industrial data augmentation framework, Real-Adapter, for generating high-quality data with complex scenarios and fine-grained local defects under limited data.
- 2) We design lightweight adapters, SIA and LLIA, to capture industrial scenarios and detailed local defect features, aligning domain-specific control signals with the foundation model. In addition, we introduce a Task-Aware (TA) loss to explicitly guide the generative process toward improving downstream defect detection performance.
- 3) We develop an industrial defect images acquisition equipment to automatically collect high-quality data for industrial components. Upon acceptance, two datasets, one with impeller surface defect data and another with cylinder surface defect data, will be available on GitHub (<https://github.com/kecheng-smile/dataset-ID-and-CBD>) for algorithm evaluation and benchmark dataset development.

2. Related work

In this section, we discuss research related to generative models and their application in industrial data augmentation.

2.1. Industrial image data augmentation

In visual manufacturing quality inspection, data-driven deep learning algorithms are widely employed due to their proven efficacy (Fu et al., 2025; Li et al., 2025). However, these methods typically require extensive datasets, rendering data scarcity a critical limitation. To mitigate this issue, image data augmentation techniques have gained prominence, as they reduce industrial data acquisition costs and improve the distribution of data samples. These techniques can be broadly classified into traditional and generative approaches. Traditional methods expand datasets through basic geometric transformations, such as rotation and scaling. Nevertheless, the samples generated by these methods often lack diversity and fail to enhance the model's generalization capability.

With the advent of deep-learning-based generative models (Zhang et al., 2021; Zheng et al., 2022), many researchers have adopted these techniques to mitigate data scarcity challenges in manufacturing quality control. For instance, Zhang et al. (2022) developed DTD-GAN, which incorporates an auxiliary classifier to enhance the diversity of generated tire defect samples by leveraging the properties of the latent space. Liu et al. (2023a) proposed Anomaly-GAN with a cycle-consistent and anomaly-aware mechanism for controllable, high-fidelity anomaly generation to enhance anomaly detection performance. Kim et al. (2024) developed an SDGAN-based framework for high-fidelity data generation in fault diagnosis, effectively mitigating training instability and mode collapse issues inherent in generative adversarial architectures. Yun et al. (2020) introduced a conditional Variational Autoencoder (VAE) approach that synthesizes sufficient metal surface defect samples for training, consequently enhancing the performance of detection models. More recently, Low et al. (2024) augmented data by converting

time-series signals into images and utilizing StyleGAN. They further employed a lifting algorithm to select optimal classifiers and optimized hyperparameters via Bayesian methods to improve detection performance. Lin et al. (2024b) proposed a strategy for analyzing inter-class imbalance in small-sample scenarios and introduced ShcNet, a lightweight classification network designed for imbalanced industrial datasets with limited samples. Collectively, these data augmentation methods partially alleviate the issue of data scarcity in production quality control.

Nonetheless, these methods encounter challenges in generating defects for industrial components characterized by complex surface structures and textures, despite their success with simpler defect types. Furthermore, the structural limitations of these models, as shown in Table 1, often result in generated data with insufficient diversity, which constrains their effectiveness for downstream tasks (Yang et al., 2024). These limitations impede the practical deployment of data augmentation techniques in manufacturing quality inspection.

In recent years, the Denoising Diffusion Probabilistic Model (DDPM) (Ho et al., 2020) has emerged as a powerful generative paradigm, demonstrating remarkable success in image synthesis. These models have surpassed GAN- and VAE-based approaches in terms of visual fidelity, controllability, and diversity, achieving state-of-the-art performance in natural image generation (Azizi et al., 2023). Concurrently, their application in industrial defect generation has seen rapid development. For instance, Zhang et al. (2025) proposed Defect-LoRA, which utilizes dual low-rank adaptations to respectively maintain carrier morphology and embed defect prototypes for controllable defect augmentation. Similarly, Wei et al. (2025) introduced a two-stage diffusion framework that disentangles normal product generation and product-agnostic defect operations. While these approaches demonstrate profound potential for industrial augmentation, they typically require substantial tunable parameters and rely on general industrial datasets for effective training. Consequently, generating high-quality, domain-specific data under few-shot conditions remains challenging. Furthermore, explicitly ensuring that the generated fine-grained defects genuinely benefit downstream detection algorithms requires further exploration. Therefore, developing highly efficient and task-aligned diffusion models remains a valuable pursuit.

This study aims to harness the advanced generative capabilities of diffusion models for industrial image synthesis. To this end, we design lightweight adapters to control the foundation model, facilitating the generation of high-fidelity and diverse industrial data that enhances the performance of downstream detection tasks.

2.2. Generative model based on deep learning

Deep learning-based image generation technology originated with Generative Adversarial Network (GAN) (Goodfellow et al., 2014), which employs a generator and a discriminator to learn data distributions and produce images. Subsequent research has extensively explored generating images of various styles through both conditional and unconditional approaches (Karras et al., 2021; Wang et al., 2018; Zhang et al., 2021). For instance, Arjovsky et al. (2017) designed Wasserstein GAN to improve image generation quality by introducing Wasserstein distance as the loss function. Building on Conditional GAN (Mirza & Osindero, 2014), Liu et al. (2019) proposed a Multi-Stage GAN to directly synthesize and inject defects into defect-free images. Karras et al. (2021) introduced StyleGAN series models (StyleGAN, StyleGAN2, and StyleGAN3), which use a mapping network to project noise into latent codes; these codes are then injected into the generator to control the style of the output image. Beyond GAN-based models, VAE-based models have also been applied to image generation. For example, Huang et al. (2023) proposed a Dynamic Quantization VAE to achieve fine-grained natural image generation. Collectively, these methods have demonstrated considerable success in generating natural images.

Recently, diffusion-based generative models have emerged due to their powerful performance in image synthesis. Substantial research in-

dicates that large foundational diffusion models, based on the DDPM, surpass GAN- and VAE-based models in visual fidelity, controllability, and diversity, achieving state-of-the-art results. Furthermore, these models avoid the mode collapse problem commonly associated with GAN training (Azizi et al., 2023). Consequently, diffusion models have become a mainstream solution for image generation. For example, Yang et al. (2024) proposed ConPreDiff, a diffusion-based method that uses context prediction to enhance image synthesis, achieving state-of-the-art text-to-image results on the MS-COCO dataset (Lin et al., 2014). Similarly, Zheng et al. (2023) introduced Layout Diffusion, which fine-tunes latent diffusion models for layout-conditioned image generation, producing high-quality outputs with lower computational cost. Zhou et al. (2024) designed a Multi-Instance Generation Controller that enhances multi-instance image generation via instance-aware attention mechanisms. To provide versatile generation control without full-model re-training, recent studies have extensively explored adapter-based mechanisms. For instance, Ye et al. (2023) introduced IP-Adapter, which enables effective image prompt capability for pre-trained diffusion models via a cross-attention mechanism. Lin et al. (2024a) proposed CTRL-Adapter, an efficient framework that adapts diverse spatial controls to various foundation models by addressing feature space mismatches. Furthermore, Zhong et al. (2025) developed Mod-Adapter, a tuning-free approach that achieves multi-concept personalization by elegantly modulating abstract and concrete features.

Despite their outstanding performance on natural images, most existing diffusion-based generative and control methods are not tailored for industrial data augmentation tasks. These approaches primarily focus on generic geometric or semantic guidance (e.g., layouts, instances, or textual descriptions) and typically require large-scale paired data for effective fine-tuning. When applied to industrial domains characterized by complex structures, strong domain-specific patterns, and extremely limited defective samples, they often struggle to capture industrial semantics and fail to reproduce accurate fine-grained defect details. Consequently, directly applying existing diffusion models to industrial data augmentation remains challenging for supporting downstream inspection tasks. To address these limitations, we propose Real-Adapter, which explicitly targets industrial data augmentation under data-scarce conditions. As shown in Table 1, unlike GAN-based data augmentation techniques, the Real-Adapter incorporates a large generative foundation model, which provides robust prior knowledge. Different from existing diffusion-based methods, Real-Adapter disentangles global industrial scene adaptation and localized defect-aware control through two lightweight and decoupled adapters, namely the SIA and LLIA branches. Furthermore, we introduce the TA loss to align the generative process with downstream defect detection objectives, which is rarely considered in prior diffusion-based image generation and augmentation frameworks.

3. Proposed method

In this section, we introduce the overall framework of the proposed Real-Adapter.

3.1. Preliminaries

Diffusion models generate high-fidelity images through a forward noise-adding Markovian process and a reverse denoising process. Given a data sample from the original distribution, the forward process gradually adds Gaussian noise over t time steps, eventually transforming the image into pure noise. During the reverse process, a neural network (typically a U-Net) is trained to predict and progressively remove the added noise at each time step. Following the standard formulation of the DDPM, the foundation model is optimized by minimizing the mean squared error between the actual noise ϵ and the predicted noise ϵ_θ at time step t :

$$L_\theta = \|\epsilon - \epsilon_\theta(x_t, t)\|^2, \quad (1)$$

Table 1

Comparison of existing generative paradigms and Real-Adapter across critical dimensions for industrial data augmentation.

Generative Paradigm	Representative Methods	Large-scale Prior Knowledge	Global Scenario Control Signal	Detailed Local Control Signal	Downstream Task-Awareness
GAN/VAE-based	StyleGAN3, DQ-VAE	×	×	×	×
Industrial Augmentation	EID-GAN, Mask2Defect	×	×	✓	×
Universal Diffusion-based	ControlNet, IP-Adapter, OminiControl	✓	×	×	×
Ours	Real-Adapter	✓	✓ (SIA branch)	✓ (LLIA branch)	✓ (TA Loss)

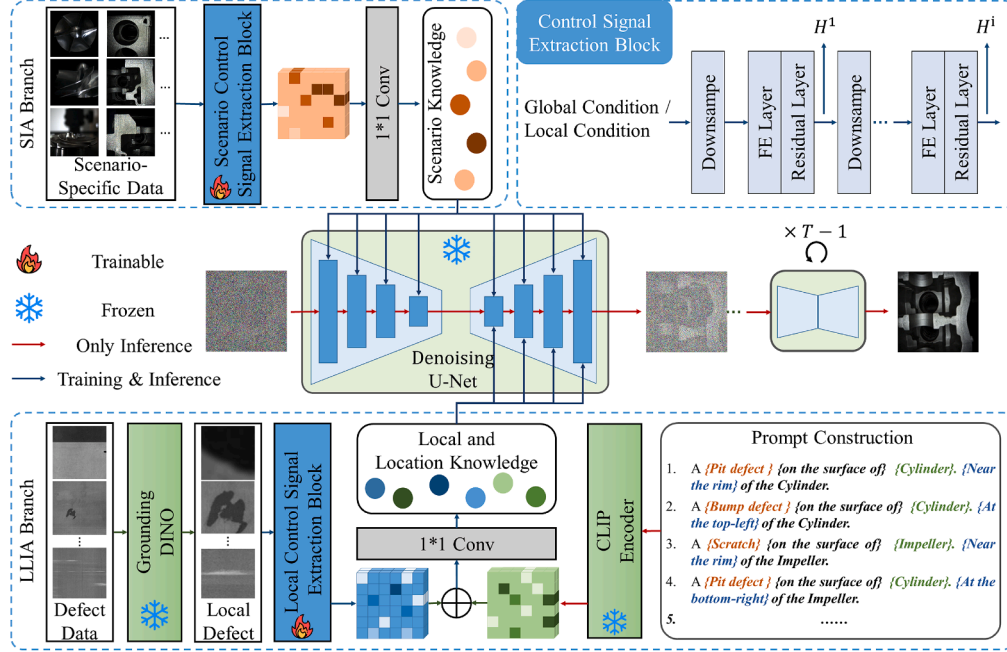


Fig. 2. The overall structure of Real-Adapter, which is designed for general industrial defect inspection data augmentation. This framework consists of two lightweight adapters, termed SIA and LLIA. These branches extract global domain-specific information of industrial scenarios, alongside fine-grained details of defect geometries and spatial localizations. The model learns to align these two control signals with a large generative foundation model, thereby enabling the generation of high-fidelity, physically plausible industrial scene data to improve downstream task performance.

Here, ϵ represents the noise that the U-Net aims to estimate. The diffusion model's loss function reveals that its core objective is to predict the noise integrated at each step. By minimizing the error between the predicted and actual noise, the diffusion model gains the capability to reconstruct the original image from noise.

3.2. Overall framework of the proposed method

As shown in Fig. 2, the overall architecture of Real-Adapter consists of three main components: (1) a large generative foundation model, specifically the denoising U-Net from Stable Diffusion v1.5; (2) an SIA branch; and (3) an LLIA branch. The proposed SIA branch receives the domain-specific images as input and leverages a scenario control signal extraction block to extract information on specific industrial scenarios. The LLIA branch integrates Grounding DINO (Liu et al., 2023b), Contrastive Language-Image Pre-Training (CLIP) Text Encoder (Radford et al., 2021), and a designed local control signal extraction block to capture local and localization information. The Grounding DINO is implemented to leverage its open-vocabulary object detection capabilities to obtain general industrial defect images from industrial defect datasets like NEU-SEG (Dong et al., 2019) and Magnetic Tile (MT) (Huang et al., 2020) datasets. By directly providing defect category names (e.g., “scratch”, “pit”) as text prompts, Grounding DINO automatically generates bounding boxes, which subsequently serve as inputs for the local control signal extraction block in the LLIA branch. The local control signal extraction block captures the detailed features of the general industrial defect output by Grounding DINO. The CLIP Text Encoder receives the text prompt as input and captures the location information of cer-

tain defects and parts. The text prompts are structured following the format: <defect> <part> <location information>. Specifically, we define five types of location cues for the ID dataset and six for the CBD dataset. All prompts remain fixed throughout the training and inference phases. The information from two branches is embedded into the denoising U-Net with the decoupled cross-attention mechanism to control the diffusion model generating industrial defect data. The overall structure is optimized by a designed TA loss for more powerful data augmentation ability. During the training process, both Grounding DINO and the CLIP Text Encoder are frozen, while the parameters of the adapters (SIA and LLIA branches) are optimized.

It is worth noting that Real-Adapter does not aim to introduce another generic diffusion model. Instead, it targets industrial defect data augmentation under extreme data scarcity, where controllability, parameter efficiency, and downstream task alignment are jointly required.

3.3. SIA, and LLIA branch

The SIA branch utilizes a scenario control signal extraction block to capture the global scenario information. As shown in the upper right corner of Fig. 2, the block comprises feature extraction (FE) layers and downsampling layers to extract scenario information. Specifically, scenario information can be expressed as follows:

$$H_{i+1,sc}^{w \times h} = \theta_{res} \left(\text{Vim}(\theta_{ds}(H_{i,sc}^{w \times h} * \frac{1}{2^{i-1}})) \right), \quad (2)$$

$$F_{sc} = K_{1 \times 1} * H_{d_{SIA,sc}}, \quad (3)$$

where $H_{i,sc}$ denotes the hidden scenario feature extracted by FE and residual layers in the scenario control signal extraction block, $i = 1, 2, \dots, d_{SIA}$. The parameter d_{SIA} denotes the depth of the SIA branch. As a critical hyperparameter, it governs the quality of the embedded scene information. In this study, d_{SIA} is set to 4. We analyze the impact of varying this parameter on model performance in our hyperparameter experiments. w and h are the width and height of the input image. $\emptyset_{res}(\cdot)$ means the residual block and $\emptyset_{ds}(\cdot)$ represents the double downsampling. $Vim(\cdot)$ denotes the Vision Mamba (Zhu et al., 2024) mechanism to extract the scenario feature, strategically leveraging state-space modeling to capture long-range dependencies across complex industrial backgrounds with a lightweight, linear computational complexity. $K_{1 \times 1}$ denotes the convolution kernel of size 1×1 and $*$ is the convolution. F_{sc} is the scenario feature extracted by the SIA branch.

To control the foundation model to generate data with fine-grained defects, we design the LLIA branch to extract the local defect feature. We first utilize the Grounding DINO to obtain the local detailed defect image patch. The local control signal extraction block subsequently extracts the industrial defect feature by utilizing the same structure as the scenario control signal extraction block. To obtain the location of defects and parts, we utilize the CLIP text encoder to encode the text input and obtain the location information. The local and location information extraction process can be expressed as follows:

$$H_{i+1,lc}^{\frac{w}{2^i} \times \frac{h}{2^i}} = \emptyset_{res} \left(\text{Atten}(\emptyset_{ds}(H_{i,lc}^{\frac{w}{2^{i-1}} \times \frac{h}{2^{i-1}}})) \right), \quad (4)$$

$$H_l = CLIP(text), \quad (5)$$

$$F_{lc} = K_{1 \times 1} * (H_{d_{LLIA},lc} \oplus H_l), \quad (6)$$

where $H_{i,lc}$ denotes the hidden local defect feature extracted by FE and the residual layers in the local control signal extraction block. Here, $i = 1, 2, \dots, d_{LLIA}$, and this parameter is critical for preserving fine-grained defect information. In this study, we set $d_{LLIA} = 3$ and investigate the impact of this value on model performance. $\text{Atten}(\cdot)$ represents the attention mechanism to extract the defect feature. H_l denotes the location information extracted by the CLIP encoder, and F_{lc} means the final local and location feature extracted by the LLIA branch.

After extracting the scenario, local defect, and localization information, we use a cross-attention mechanism to introduce these control signals into the foundational denoising U-Net. Scenario information is integrated during both the encoder and decoder of U-Net, whereas local defect and location knowledge are incorporated only in the decoder. The control signal fusion process in the U-Net encoder can be expressed as follows:

$$F'_{en,i} = \text{Atten}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V, \quad (7)$$

where $Q = F_{en,i} \cdot W_q$, $K = F_{sc} \cdot W_k$, $V = F_{sc} \cdot W_v$ are the query, key, and value matrices of the attention operation respectively, and W_q , W_k , and W_v are the weight matrices. $F_{en,i}$ and $F'_{en,i}$ denote the input and output of cross attention in the i th layer of encoder.

In the decoder control signal fusion process, a decoupled cross-attention mechanism is used to integrate the two control signals into U-Net more effectively. The calculation can be expressed as follows:

$$F'_{de,i} = \text{Atten}(Q, K, V) \oplus \text{Atten}(Q, K', V'), \quad (8)$$

where $K = F_{lc} \cdot W_k$ and $V = F_{lc} \cdot W_v$ are the key and value matrices used in cross-attention with features from the LLIA branch. Similarly, $K' = F_{sc} \cdot W'_k$ and $V' = F_{sc} \cdot W'_v$ are the key and value matrices used in cross-attention with features from the SIA branch. The same query $Q = F_{de,i} \cdot W_q$ is used for both local control signal and scenario control signal cross-attention. Finally, the output of the cross-attention in each decoder layer is obtained by summing the results of two decoupled cross-attention operations.

Analysis of Control Signal Fusion Methods: By integrating the SIA and LLIA branches, we incorporate scene-level knowledge and fine-grained defect cues into a large-scale foundation generative model, thereby enabling controlled image synthesis. SIA is injected into both the encoder and decoder blocks to guide the emergence of global structure and texture statistics in the early denoising trajectory, which is critical for industrial realism. In contrast, LLIA is injected primarily into the decoder blocks, because fine-grained defect geometry and boundary details are mainly refined during the decoder process. Moreover, injecting LLIA too early could distort global structures and exacerbate artifacts.

$$F'_{en,i} = \text{Atten}(Q, K, V) \oplus \text{Atten}(Q, K', V'), \quad (9)$$

As shown in Eq. (9), all control signals from LLIA and SIA are injected into the encoder layers of the diffusion U-Net. This fusion strategy may distort the global image structure. The subsequent ablation studies and visualizations further demonstrate the limitations of this fusion strategy.

In addition, when SIA and LLIA are inserted simultaneously at the decoder, we compute separate attention distributions for each control signal. This design is motivated by the fact that scenario-level controls (SIA) are global and dense, whereas defect-level controls (LLIA) are sparse and highly localized. When these heterogeneous signals are fused in a coupled manner, as described below, they share a single softmax normalization, which can induce feature conflicts: dominant scenario tokens may capture most of the attention mass, thereby suppressing defect tokens and attenuating defect-related gradients.

$$F'_{de,i} = \text{Softmax} \left(\frac{Q(K \oplus K')}{\sqrt{d}} \right) (V \oplus V'), \quad (10)$$

$$F'''_{de,i} = \text{Softmax} \left(\frac{Q(\text{Concat}(K, K'))}{\sqrt{d}} \right) (\text{Concat}(V, V')), \quad (11)$$

As shown in Eqs. (10) and (11), we couple heterogeneous control signals under a shared softmax normalization and then inject the coupled signal into the decoder layers of the diffusion model. This design hampers the preservation of fine-grained defect cues, particularly under complex background conditions.

Overall, injecting defect cues at encoder stages or performing unified fusion tends to cause (i) defect dilution (blurred or missing defects), (ii) location drift (defects deviate from their target positions), and (iii) background artifacts (unnatural textures on industrial surfaces), especially under complex industrial backgrounds. We further evaluate and compare the effectiveness of various control-signal fusion strategies in our experiments.

Comparison with LoRA-style tuning: LoRA is a widely used parameter-efficient tuning strategy for diffusion models, which injects low-rank updates into the backbone weights (typically in attention projections) without updating the full model. Given a projection matrix $W \in \mathbb{R}^{d_{in} \times d_{out}}$ (e.g., W_Q, W_K, W_V), LoRA parameterizes the update as follows:

$$W' = W + \Delta W, \quad \Delta W = BA, \quad B \in \mathbb{R}^{d_{in} \times r}, \quad A \in \mathbb{R}^{r \times d_{out}}, \quad (12)$$

where $r \ll \min(d_{in}, d_{out})$ is the rank. Consequently, the attention computations become $Q = HW'_Q$, $K = HW'_K$, and $V = HW'_V$, implying that LoRA adapts the backbone attention weights to fit a target domain.

In contrast, the Real-Adapter keeps the diffusion backbone frozen and introduces two independent lightweight branches that provide external control signal with (i) global industrial semantics (SIA) and (ii) localized defect feature (LLIA). These heterogeneous controls are fused via decoupled cross-attention (Eq. (8)), which proves advantageous in data-scarce industrial settings. Rather than entangling scenario-level and defect-level objectives within the same low-rank updates of the backbone, this design encourages specialization (global realism versus local defect fidelity) and improves controllability without altering the pre-trained generative priors. We note that although LoRA is effective for generic adaptation, the Real-Adapter is tailored for the cross-domain, multi-granularity control with limited data required in industrial defect augmentation.

3.4. Task-aware loss

In diffusion-based image generation, the standard training objective is to predict the noise removed at each denoising step, as formalized in Eq. (1). This loss function guides the model in the iterative denoising process to reconstruct the final image. However, for the specific task of generating industrial images to augment downstream detection models, this objective lacks explicit constraints for industrial task awareness, limiting its efficacy for targeted data augmentation. To enhance the model's utility for downstream detection performance, we incorporate task aware constraints during the training process to ensure that the downstream detection model correctly identifies the generated defects. This approach ensures that the generated defects align more strictly with physical principles, and their feature distributions closely match those of actual defects.

Specifically, during the diffusion process, the Segment Anything Model (SAM) applies semantic segmentation to noisy images, producing masks as Ground Truth. In the reverse denoising process of the training stage, SAM also segments defects in the generated images. The generative model is optimized using the loss between the segmentation output of the denoised image and its corresponding mask. This ensures that the generated images, depicting industrial parts or defects, can be accurately segmented by the pre-trained large vision model, similar to real images. This downstream task-guided training loss enhances the model's ability to generate high-quality augmented data for downstream tasks. The TA loss can be expressed as:

$$L_{TA} = \alpha L_{dice} + (1 - \alpha) L_{boundary} \quad (13)$$

where L_{TA} denotes the TA loss, L_{dice} and $L_{boundary}$ denotes dice loss and boundary loss respectively. α represents the weight balance of L_D and L_B . In our hyperparameter experiments, we investigate the impact of this value on model performance. L_D and L_B can be expressed as follows:

$$L_{dice} = 1 - \frac{1}{cla} \sum_{i=1}^{cla} \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} \quad (14)$$

$$L_{boundary} = \frac{1}{cla} \sum_{cla=1}^{cla} \sum_{num=1}^{num} \delta(g_i^{cla}) |p_i^{cla} - g_i^{cla}| \quad (15)$$

where cla and num represent the number of classes and pixels, respectively. p_i and g_i represent the predicted values from the denoising process and the true values of the mask labeled by the SAM, respectively. $\delta(\cdot)$ is the weight function, expressed as follows:

$$\delta(g(x)) = \begin{cases} 1, & \text{if } |g(x)| < \epsilon \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

The final training loss of Real-Adapter can be formulated as follows:

$$\mathcal{L} = \beta L_\theta + (1 - \beta) L_{TA} \quad (17)$$

where β denotes the balancing weight between L_θ and L_{TA} .

From the optimization perspective, the standard diffusion objective mainly contributes pixel-level denoising gradients that drive the synthesized data toward the target statistical distribution. However, such pixel-wise supervision does not explicitly constrain the semantic completeness of defect regions or the sharpness of their physical boundaries. By introducing the TA loss, the overall gradient flow can be written as:

$$\nabla_\theta \mathcal{L} = \beta \nabla_\theta L_\theta + (1 - \beta) \nabla_\theta L_{TA} \quad (18)$$

where $\nabla_\theta L_{TA}$ serves as an additional semantic- and boundary-aware supervisory signal that complements the original denoising gradient. It should be noted that SAM is used only as a frozen pseudo-label generator, and the pseudo-masks are treated as detached supervision targets. Therefore, the gradients induced by L_{TA} do not propagate through SAM, but are back-propagated through the denoised prediction and then into

the diffusion model parameters. Consequently, the TA loss explicitly encourages the model to generate defects with more complete topology and clearer physical boundaries.

During training, the balancing weight β is designed to decay gradually throughout the training process. This schedule is motivated by the observation that the model initially focuses on learning the global image layout during the early training stage and then progressively emphasizes fine-grained defect refinement in the later stage. Therefore, we introduce a decay coefficient d_β for this loss weight. The initial value of β is set to 0.9, with d_β set to 0.05, implementing a linear decay schedule in which β is reduced by 0.05 after every 1×10^4 training iterations. The impact of these hyperparameters on model performance is analyzed in our ablation studies.

To reduce the risk of segmentation instability caused by diffusion noise, pseudo-labels are generated only in the early noise-addition stages (i.e., the first three steps), and the TA loss is computed in the corresponding denoising stages. As demonstrated in our supplementary materials, when the diffusion step is below 5, SAM's segmentation performance remains largely unaffected by noise and can serve as a relatively robust segmentation tool. Concurrently, this design acknowledges that the generative process refines detailed information in the later stages. By applying task constraints during these final denoising step, we enable the model to focus on refining local defect details as the generation process nears completion.

When using the TA loss, it is also necessary to consider potential failure modes caused by the segmentation performance of SAM. For example, if the pseudo-labels generated by SAM are under-segmented or over-segmented, the TA loss may bias optimization toward incomplete defects or encourage the model to synthesize spurious artifacts in irrelevant regions. In addition, if SAM fails to produce reliable segmentation results on generated images, the task-aware supervision may become unstable and introduce biased gradients during optimization. To mitigate potential errors in TA loss calculation arising from the inherent limitations of SAM, we first exclude samples lacking stable segmentation results from TA-loss supervision, ensuring that unreliable supervision does not dominate optimization. Specifically, for each training sample, SAM masks are generated over the selected early noising steps. If the segmentation result are empty or exhibit large fluctuations across these steps, this sample is discarded from the training process. This approach filters out samples containing defects that SAM fails to segment. Thus, during training, if SAM fails to segment a generated defect, we can infer that the defect exhibits a substantial discrepancy in shape, position, or intrinsic feature distribution compared to the defects in the original training samples, rendering it unsegmentable. Although this approach penalizes some seemingly reasonable generated defects and reduces the diversity of the generated outputs, the task constraint ensures that the generated samples conform to the distribution of real defect features and improves compatibility with downstream tasks. To further ensure the stability of the TA loss, we average the segmentation loss across the diffusion steps, providing an additional safeguard against computational instability. By employing the aforementioned strategies, we effectively mitigate the impact of SAM's failure modes on the computation of the TA loss. The experimental results further validate the effectiveness of this loss in improving downstream task performance.

4. Experiments

This section introduces the data collection equipment, datasets and the quantitative analysis of Real-Adapter.

4.1. Dataset construction

To construct real industrial scene datasets for verifying the effectiveness of our method, we construct a flexible industrial part surface quality data acquisition equipment. This equipment mainly includes a

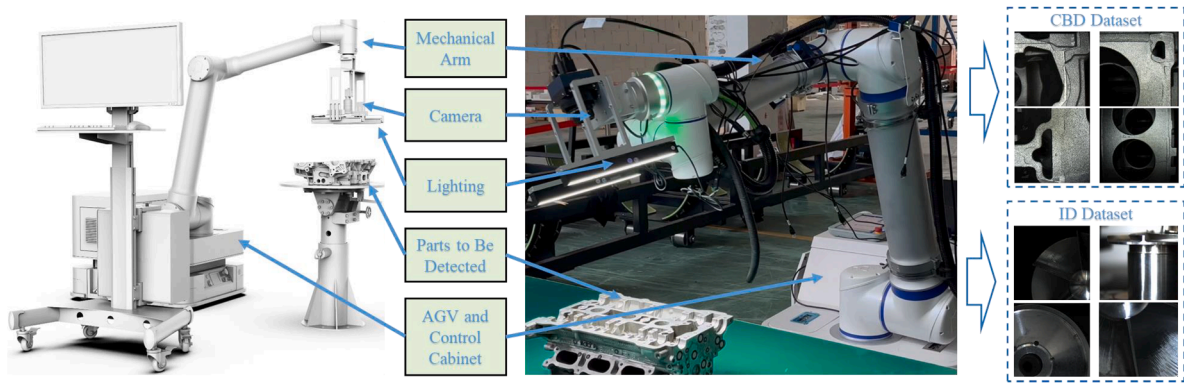


Fig. 3. Configuration of the flexible industrial part surface defect data acquisition equipment. This equipment consists of a six-axis collaborative mechanical arm, a charge-coupled device (CCD) camera, two linear lighting sources, and an AGV. We utilize this system to automatically collect defect data from the CBD and ID datasets.

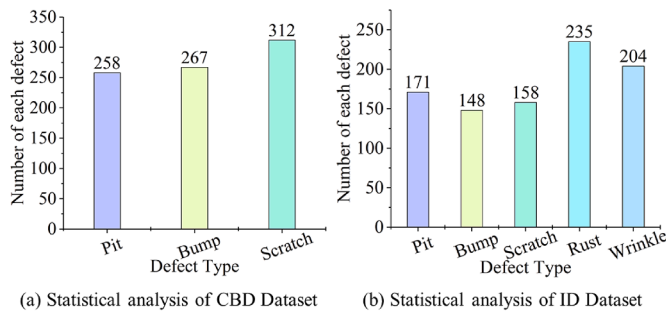


Fig. 4. A detailed statistical analysis of the CBD and ID datasets, which demonstrates the balanced distribution of defects within each collection.

Charge-Coupled Device (CCD) Camera, Lighting, a Six-Axis Collaborative Mechanical Arm, an AGV, and a control cabinet, as shown in Fig. 3.

Using this equipment, we construct two real industrial datasets, the Cylinder Block Defect (CBD) dataset, and the Impeller Defect (ID) dataset, each comprising 342 and 354 surface defect image samples from industrial parts. The signs of various defects are shown in Table 2 and the samples in these two datasets are shown in Fig. 3.

CBD dataset: The dataset contains 342 cylinder block defect images of size 512×512 with three types of defects: Pit, Bump, and Scratch. As illustrated in Fig. 4(a), the dataset includes 258, 267, and 312 instances for each respective defect type, which indicates a relatively balanced distribution of targets. A total of 300 samples are employed for training the generative models, while the remaining images serve as a validation set to prompt the models for image generation.

ID dataset: The dataset contains 354 impeller defect images of size 512×512 with five types of defects: Pit, Bump, Scratch, Rust, and Wrinkle. As illustrated in Fig. 4(b), the dataset includes 171, 148, 158, 235, and 204 instances for each respective defect type, which indicates a relatively balanced distribution of targets. A total of 300 samples are utilized for training the generative models, while the remaining 54 images constitute the validation set, which is used to prompt the models for image generation.

To validate the generalization capability of Real-Adapter, we select two public industrial surface defect datasets, NEU-SEG and MT, for further generalization experiments. NEU-SEG dataset focuses on hot-rolled steel strip surface defects, such as patches, scratches, and inclusions, with each image sized at 200×200 pixels. MT dataset consists of 1344 images with 459 labeled objects belonging to 5 different classes, including blowhole, uneven, break, crack, and fray. For each public dataset, 300 annotated images are selected for training, and 50 samples are reserved for validation.

Table 2

Visualization of various defects within the datasets and descriptions of their physical characteristics.

Defect Class	Defect Feature	Sample Images
Wrinkle	Visible wavy or irregular patterns.	
Rust	Reddish-brown patches or spots on the metal surface.	
Bump	Irregular shapes ranging from small to larger deformations.	
Pit	Small depressions or cavities on the surface of the metal.	
Scratch	Slender and irregular marks on the metal surface.	

4.2. Implementation details

The experimental hardware configuration comprises four NVIDIA RTX 4090 GPUs, each equipped with 24 GB of memory. For the diffusion-based models, we employ the publicly available Stable Diffusion v1.5 as the pre-trained backbone. The models are trained for 1×10^5 iterations with a batch size of 1 per iteration. Optimization is performed using the AdamW optimizer with a learning rate of 1×10^{-4} . During inference, the denoising process utilizes 50 sampling steps. For other generative methods evaluated in this study, we meticulously optimized the hyperparameters to maximize performance. This ensures an unbiased evaluation, and all baseline models are trained until convergence under the identical constraint of limited training samples.

Additionally, we employ Grounding DINO 1.5 Pro to extract defect image patches and set both the box and text thresholds to 0.25. For the automatically cropped image patches, we conduct a subsequent manual verification of the patches to ensure strict accuracy. Furthermore, we provide a calibration experiment in the supplementary material to validate the reliability of CLIP and Grounding DINO.

Table 3

Quantitative comparison of Real-Adapter with representative generative methods on ID and CBD.

Methods	Backbone	ID					CBD				
		FID(↓)	KID×10 ³ (↓)	IS(↑)	LPIPS(↓)	DS(↑)	FID(↓)	KID×10 ³ (↓)	IS(↑)	LPIPS(↓)	DS(↑)
EID-GAN ⁱ	U-Net	97.541	33.412	6.573	0.392	0.341	92.133	31.542	7.524	0.378	0.376
DLS-GAN ⁱ	U-Net	94.064	31.854	5.782	0.400	0.210	103.742	36.218	6.485	0.478	0.342
Mask2Defect ⁱ	U-Net	99.026	34.215	6.241	0.442	0.330	91.237	30.875	7.239	0.429	0.356
Pix2pixHD ^g	U-Net	146.474	58.741	3.332	0.476	0.238	148.290	60.125	3.895	0.507	0.276
Defect-GAN ^g	U-Net	137.822	52.194	2.972	0.450	0.261	142.920	54.382	3.214	0.429	0.297
StyleGAN3 ^g	U-Net	109.086	38.642	6.871	0.405	0.297	104.589	35.814	6.740	0.397	0.334
VAE ^v	Autoencoder	182.645	75.321	2.973	0.428	0.312	177.990	73.145	3.107	0.478	0.320
MoVQ ^v	Autoencoder	87.822	28.163	6.762	0.407	0.429	89.231	29.541	6.599	0.270	0.419
DQ-VAE ^v	Autoencoder	92.033	30.578	6.836	0.356	0.472	82.989	26.837	7.516	0.253	0.428
ControlNet ^d	Stable Diffusion v1.5	95.679	33.105	6.020	0.389	0.300	102.691	34.912	6.751	0.316	0.327
T2I ^d	Stable Diffusion v1.5	79.652	26.421	6.575	0.283	<i>0.442</i>	68.956	21.563	7.437	<i>0.254</i>	0.408
OminiControl ^d	Stable Diffusion v1.5	<u>77.821</u>	<u>24.238</u>	<i>7.038</i>	0.299	0.412	79.652	25.148	7.895	0.274	0.417
IP-Adapter ^d	Stable Diffusion v1.5	61.956	18.765	<i>7.153</i>	0.276	0.389	57.232	16.892	<i>7.627</i>	<i>0.252</i>	0.422
Ours	Stable Diffusion v1.5	45.122	<u>11.342</u>	7.836	0.205	0.378	42.597	<u>10.453</u>	8.152	0.200	0.423

Note: The methods with the *i*, *g*, *v*, and *d* denote industrial data augmentation methods, GAN-based generative models, VAE-based models, and diffusion-based models, respectively. Bold, Italic, and Underline symbolize the top-three performance of each metric.

4.3. Evaluation metrics

In these experiments, we evaluate the proposed Real-Adapter from two primary perspectives: the quality of the generated industrial images and their effectiveness in augmenting various downstream visual inspection tasks.

To assess generation quality, we employ four standard metrics: Fréchet Inception Distance (FID), Kernel Inception Distance (KID), Inception Score (IS), Learned Perceptual Image Patch Similarity (LPIPS), and Diversity Score (DS). Specifically, FID, KID and LPIPS measure the authenticity and visual fidelity of the generated images by quantifying their distributional and perceptual similarity to real industrial data. High-quality image generation is a prerequisite for effective data augmentation. IS and DS evaluate the generative diversity, ensuring that the augmented datasets encompass a balanced distribution of defect morphologies and prevent mode collapse.

To comprehensively evaluate the augmentation effectiveness across different industrial inspection requirements, we establish a rigorous evaluation protocol encompassing both semantic segmentation and instance-level localization tasks:

To validate the model's ability to enhance pixel-wise defect classification, we adopt representative semantic segmentation architectures called DeepLabv3+ (Chen et al., 2018). These models are trained using samples synthesized by various generative methods, allowing us to evaluate the specific augmentation improvements brought by each method. We employ mean Intersection-over-Union (mIoU) and Pixel-wise Mean Average Precision (mAP) scores to quantify these improvements. Furthermore, since industrial inspection often requires isolating individual defect instances and pinpointing their exact locations, we adopt an instance segmentation model, YOLOv11-seg (Khanam & Hussain, 2024). After training these models on the differently augmented datasets, we assess their enhanced localization and instance-level recognition capabilities using mAP and mIoU.

4.4. Overall quantitative analysis of Real-Adapter

In this section, we aim to comprehensively evaluate the generation and downstream task augmentation capabilities of Real-Adapter. 13 mainstream generative approaches, namely Mask2Defect (Yang et al., 2021), EID-GAN (Li et al., 2022), DLS-GAN (Li et al., 2023), Pix2pixHD (Wang et al., 2018), Defect-GAN (Zhang et al., 2021), StyleGAN3 (Karras et al., 2021), VAE (Kingma et al., 2013), MoVQ (Zheng et al., 2022), DQ-VAE (Huang et al., 2023), ControlNet (Zhang et al., 2023), IP-Adapter (Ye et al., 2023), T2I (Mou et al., 2024), OminiCon-

trol (Tan et al., 2024) are selected for comparing the effectiveness on the above datasets.

4.4.1. Comparison with mainstream models

To assess the generative performance of Real-Adapter, we compare it against 13 mainstream image generation methods, spanning industrial data augmentation, GAN-based, VAE-based, and diffusion-based models. These methods are evaluated on the ID and CBD datasets, which contain structurally complex industrial components, as shown in Table 3. Experimental results show that Real-Adapter outperforms all representative generative methods in producing high-fidelity industrial defect images. Specifically, its FID scores surpass the second-best model by 16.834 and 14.645, respectively. In addition, its KID ($\times 10^3$) scores outperform the second-best model by 7.423 and 6.439, respectively. Moreover, results indicate strong diversity, ensuring augmentation does not confine the dataset to long-tail features. IS scores exceed the second-best model by 0.683 and 0.257, respectively. While DS scores are marginally lower than some VAE- and diffusion-based approaches, Real-Adapter surpasses them in generating authentic industrial defects. This primarily stems from the feature extraction branch uniquely designed within Real-Adapter, which supplies precise control signals to the foundation model. Furthermore, owing to the lightweight architecture of the two branches, Real-Adapter achieves optimal training performance even with limited data.

GAN-based models, as representative approaches for image generation, exhibit relatively modest performance across two datasets, indicating they can not learn the relatively complex structural and textural features through limited training data. For instance, on the ID and CBD datasets, the FID scores of the best-performing GAN-based method (StyleGAN3) are 63.964 and 61.992 higher, respectively, than those of our Real-Adapter. Similarly, its KID ($\times 10^3$) scores are 27.300 and 25.361 higher, respectively. In contrast, VAE-based models perform strongly in terms of generation diversity across both datasets. For example, on the ID and CBD datasets, DQ-VAE achieves the highest Diversity Score (DS) of 0.472 and 0.428, exceeding the second-best model by 0.030 and 0.005, respectively. This strength can be attributed to their probabilistic latent space from which outputs are sampled. Nevertheless, these methods still face challenges in generating complex structures and textures, as evidenced by their high FID and KID scores.

Diffusion-based models exhibit strong performance on both datasets, indicating that the incorporated prior knowledge effectively provides structural and textural information on complex industrial parts. However, their lack of domain-specific feature extraction mechanisms renders them less effective than Real-Adapter for generating complex industrial scenarios and defects. Recent industrial data augmentation ap-

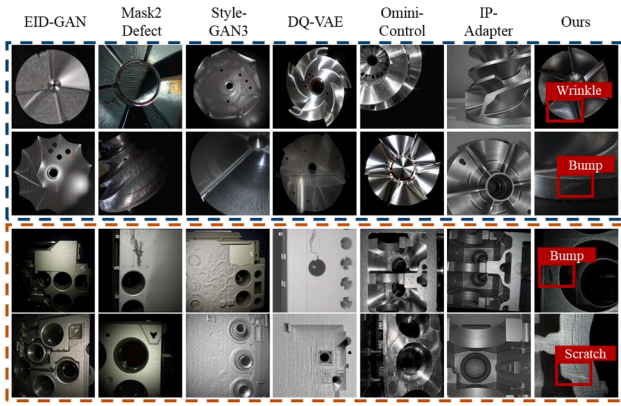


Fig. 5. Visualization results of seven leading generative methods on the ID and CBD datasets. The blue and orange dashed lines represent the generated results from the ID and CBD datasets, respectively. Compared to alternative methods, Real-Adapter generates images with more realistic textures and physically plausible defects.

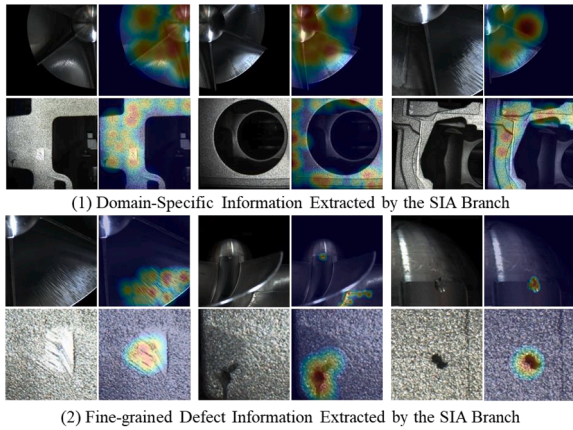


Fig. 6. Grad-CAM visualizations for the SIA and LLIA branches. These results demonstrate that the scene information and local defect details extracted by the respective branches accurately localize target regions, confirming their robust feature extraction capabilities.

proaches, such as EID-GAN and Mask2Defect, incorporate advanced data-enhancement mechanisms to generate detailed defect characteristics more effectively than GAN-based methods. Yet, their absence of prior knowledge limits their capacity to generate complex textures, resulting in inferior performance compared to diffusion-based methods.

To further demonstrate the superiority of our proposed Real-Adapter, we visualize the generative results of seven leading methods on two datasets. As shown in Fig. 5, for the ID and CBD datasets, the structure and texture of the industrial parts generated by Real-Adapter are highly realistic, and the defects (in the brown box) adhere to physical principles. The diffusion-based method also shows good performance on the overall structure of the parts, but the texture information and structure of the generated parts lack realism due to the absence of domain-specific information. Although the data generated by the industrial data-enhanced method is less effective than the diffusion-based method in terms of structure, the generated defects exhibit high realism, particularly in the detailed defect features. To ensure the integrity of our experiments, we also provide a qualitative analysis to confirm that the model does not suffer from mode collapse, which is analyzed in the supplementary materials.

To more effectively demonstrate the model's efficacy, we employ the Grad-CAM (Selvaraju et al., 2017) algorithm to visualize the industrial scene information and detailed defect features extracted by the SIA and

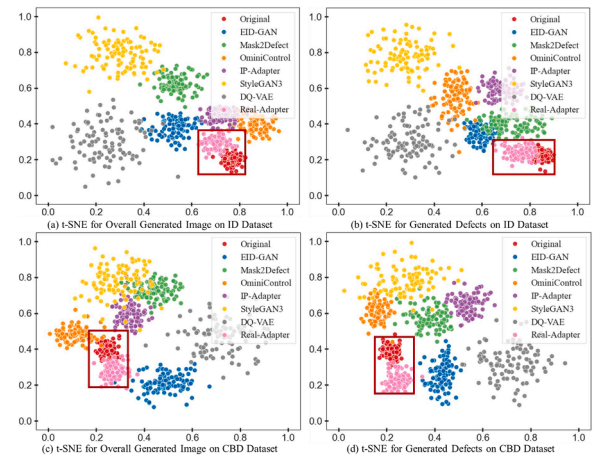


Fig. 7. Visualization of t-SNE results for generated overall images and local defects of various generative methods on ID and CBD datasets. Experimental results demonstrate that the distributions of the overall images and local defects generated by the Real-Adapter on the ID and CBD datasets closely align with those of the original images, indicating strong spatial similarity.

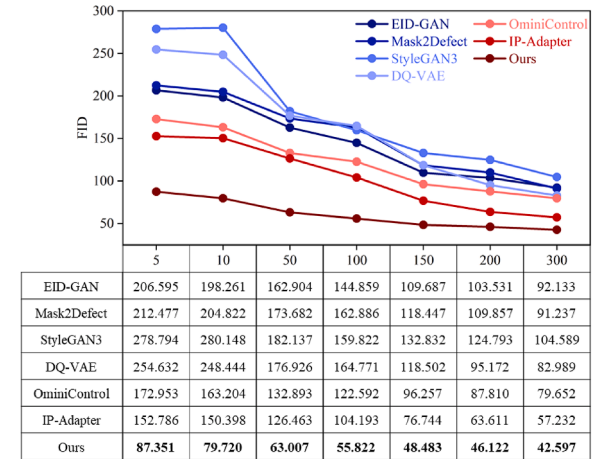


Fig. 8. Generative experiments under various numbers of training samples. The empirical result demonstrates that the lightweight architecture of Real-Adapter ensures enhanced model adaptability, enabling the framework to learn how to generate authentic industrial data under conditions of extreme data scarcity.

LLIA branches, as shown in Fig. 6. The visualization reveals that the two branches successfully focus on global surface characteristics and local defect regions of industrial parts, thereby providing robust informational support for the subsequent generative process.

The t-distributed stochastic neighbor embedding (t-SNE) images visualize the feature distributions of both generative images and local defects, as shown in Fig. 7. It can be observed that the distributions of the overall images and local defects generated by Real-Adapter (scattered points in the brown box) on ID and CBD datasets are closely aligned with those of the original images, indicating a high degree of spatial similarity. The t-SNE results for overall images obtained using OminiControl and IP-Adapter are favorable, suggesting that diffusion-based methods generate high-quality overall images owing to the strong generalization provided by prior knowledge. However, they underperform compared to industrial data augmentation methods, such as EID-GAN and Mask2Defect, in generating local detailed features, as it is difficult for diffusion-based methods to acquire detailed domain-specific knowledge, which industrial data augmentation methods can capture through few-shot training.

Table 4

The complexity and efficiency comparison of diffusion-based methods under a unified setting, demonstrating the lightweight characteristics of the proposed Real-Adapter.

Methods	Trainable Params (M)	Total Params (M)	Param Overhead (%)	GFLOPs/step (512 ²)	VRAM infer (GB)	Latency infer (s/img)	VRAM train (GB)	Train time @1e5 iters (h)
ControlNet	361.00	1427.00	33.86	910.00	8.33	5.35	21.31	23.99
T2I	77.00	1143.00	7.22	729.06	7.67	4.29	16.33	19.22
OminiControl	48.70	1114.70	4.57	711.03	7.60	4.18	15.84	18.74
IP-Adapter	22.10	1088.10	2.07	694.08	7.54	4.08	15.37	18.30
Ours	7.79	1073.79	0.73	684.96	7.50	4.03	15.12	18.06

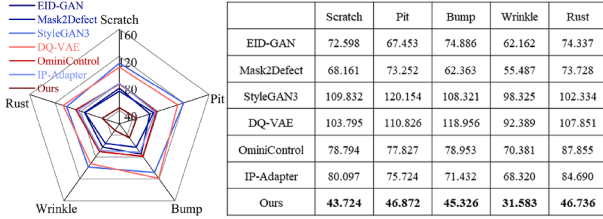


Fig. 9. Generative experiments for defects with different physical properties. The empirical evaluation demonstrates that Real-Adapter achieves robust generative performance across diverse defect categories.

To assess the generative performance of these methods under more extreme conditions of industrial data scarcity, we train seven leading models using progressively reduced training data volumes. Quantitative results for the CBD dataset are illustrated in Fig. 8. Notably, Real-Adapter achieves exceptional generative performance under extreme data scarcity, ensuring its ability to effectively leverage limited training samples to learn to precisely align extracted domain-specific features with the diffusion model's latent space, thereby generating robust synthetic outputs.

Since industrial defects exhibit distinct physical properties, such as scratches, which are typically thin and tortuous, and wrinkles, which exhibit irregular patterns, it is crucial to evaluate the model's performance in generating defects with diverse physical properties. The generative performance of five types of defects in the ID dataset, including scratch, wrinkle, bump, rust, and pit, are compared and analyzed, as shown in Fig. 9. It is evident that Real-Adapter achieves robust generative performance across all defect types.

4.4.2. Lightweight experiments

To substantiate the lightweight claim, we report the computational and efficiency metrics of diffusion-based models trained with the same backbone and hardware, including the number of trainable parameters, the trainable-parameter ratio, peak GPU memory for training and inference, inference latency, and training time, as shown in Table 4. Specifically, Real-Adapter updates only 7.79M trainable parameters. Compared with the second most parameter-efficient baseline, IP-Adapter, this reduces the trainable-parameter overhead by 1.34 percentage points. This efficiency arises because Real-Adapter employs a minimal number of layers for feature extraction and learns to align these control signals with the diffusion model, rather than updating the entire diffusion architecture. Consequently, this lightweight design enables the model to achieve optimal performance even when addressing industrial data augmentation tasks with restricted training samples. Since optimizer states and gradients scale approximately linearly with the number of trainable parameters, this compact parameterization yields the smallest training footprint, requiring 15.12 GB VRAM and saving 0.25 GB relative to the runner-up, while also reducing the extrapolated training time by 0.24 h over 10^5 iterations. For inference, where the cost is dominated by the Stable Diffusion denoising U-Net repeated for 50 steps, Real-Adapter remains closest to the backbone-level complexity: it achieves 684.96 GFLOPs per denoising step, 9.12 GFLOPs less than the next-best method, yielding the lowest latency (4.03 s/img) and the lowest peak inference

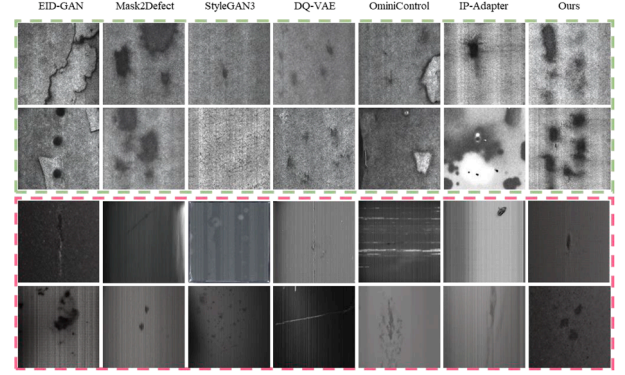


Fig. 10. Visualization results of seven leading generative methods on the NEU-SEG and MT datasets. The green and pink dashed lines denote the generated results for the NEU-SEG and MT datasets, respectively. Experimental results demonstrate that data synthesized by Real-Adapter exhibit high fidelity.

memory (7.50 GB). The inference-time margins are modest, as expected given the shared backbone, but the training-side gains are consistent and can be directly attributed to the reduced number of trainable parameters.

4.4.3. Generalization experiments

To evaluate the generalization ability of Real-Adapter, we also perform the generalization experiments on the NEU-SEG and MT datasets. The experimental results are shown in Table 5. It is evident that, compared to the ID and CBD datasets, various methods achieve relatively good generative effects on the NEU-SEG and MT datasets, as these two publicly available datasets contain only detailed defect information without complex parts or scene data. Real-Adapter slightly outperforms the second-ranked model in generative effect on these two datasets, with an FID exceeding the second-best model by 1.112 and 3.531, and the IS indicator surpassing it by 0.602 and 0.277, respectively. In addition, its KID ($\times 10^3$) scores outperform the second-best model by 0.891 and 1.069, respectively. It is worth noting that recent industrial data augmentation methods outperform diffusion-based methods on the NEU-SEG and MT datasets due to their stronger focus on detailed defect information.

A visualization of the generative results from the seven leading methods on the NEU-SEG and MT datasets is presented in Fig. 10. Due to the simple background and structure of the NEU-SEG and MT datasets, various generation models yield satisfactory results. In comparison to the diffusion-based method, both our method and industrial data augmentation methods focus on extracting industrial defect features, resulting in superior generation outcomes. The diffusion-based method, however, demonstrates mediocre performance in generating detailed defect features, as diffusion-based methods typically require larger training datasets to acquire domain-specific knowledge.

The t-SNE results for the generated images and local defects on the NEU-SEG and MT datasets are presented in Fig. 11. It is evident that the feature distributions of both overall images and local defects generated by Real-Adapter (scattered points in the brown box) are closely aligned with those of the original images. In addition, EID-GAN and Mask2De-

Table 5
Quantitative comparison of Real-Adapter with representative generative methods on NEU-SEG and MT.

Methods	Backbone	NEU-SEG					MT				
		FID($\downarrow$)	KID $\times 10^3$ ($\downarrow$)	IS($\uparrow$)	LPIPS($\downarrow$)	DS($\uparrow$)	FID($\downarrow$)	KID $\times 10^3$ ($\downarrow$)	IS($\uparrow$)	LPIPS($\downarrow$)	DS($\uparrow$)
EID-GAN ⁱ	U-Net	38.121	8.542	8.971	0.191	0.482	34.829	5.941	9.472	0.186	0.455
DLS-GAN ⁱ	U-Net	40.238	9.873	8.325	0.212	0.451	42.736	10.125	8.894	0.228	0.379
Mask2Defect ⁱ	U-Net	36.235	<u>7.015</u>	<u>8.752</u>	0.189	0.443	37.741	<u>7.325</u>	<u>9.310</u>	0.193	0.401
Pix2pixHD ^s	U-Net	46.295	13.512	5.942	0.257	0.328	52.716	16.543	5.092	0.280	0.283
Defect-GAN ^s	U-Net	42.361	11.054	7.137	0.245	0.349	47.632	13.452	7.148	0.278	0.204
StyleGAN3 ^s	U-Net	43.645	11.895	7.174	0.231	0.437	43.217	10.438	7.433	0.241	0.257
VAE ^v	Autoencoder	91.246	39.875	4.879	0.567	0.378	89.172	38.652	4.233	0.498	0.339
MoVQ ^v	Autoencoder	41.217	10.321	7.064	0.212	0.492	42.567	9.842	6.723	0.208	<u>0.471</u>
DQ-VAE ^v	Autoencoder	43.573	11.642	6.984	0.223	0.500	44.167	11.234	6.564	0.226	0.446
ControlNet ^d	Stable Diffusion v1.5	55.623	19.432	6.251	0.207	0.402	56.163	18.234	7.627	0.247	0.389
T2I ^d	Stable Diffusion v1.5	47.927	14.652	7.431	0.201	<u>0.483</u>	47.909	13.621	7.484	0.237	0.468
OminiControl ^d	Stable Diffusion v1.5	48.842	15.214	7.354	0.217	0.472	46.636	12.876	7.395	0.257	0.432
IP-Adapter ^d	Stable Diffusion v1.5	46.582	13.843	7.936	0.197	0.463	43.245	10.512	7.757	0.235	0.482
Ours	Stable Diffusion v1.5	35.123	<u>6.124</u>	9.573	<u>0.190</u>	0.469	31.298	<u>4.872</u>	9.749	<u>0.203</u>	<u>0.475</u>

Note: The methods with the *i*, *g*, *v*, and *d* denote industrial data augmentation methods, GAN-based generative models, VAE-based models, and diffusion-based models, respectively. Bold, Italic, and Underline symbolize the top-three performance of each metric.

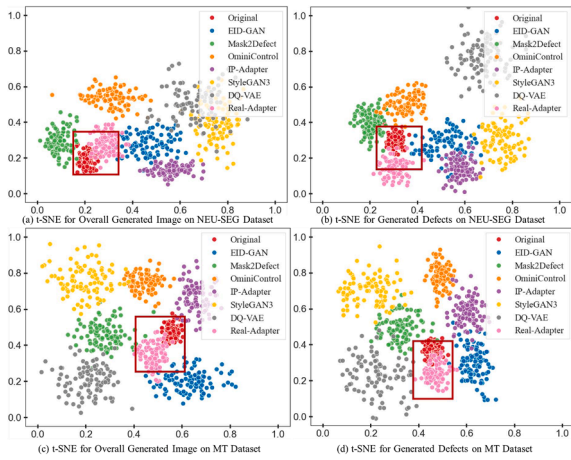


Fig. 11. Visualization of t-SNE results for generated overall images and local defects of various generative methods on NEU-SEG and MT datasets. Experimental results demonstrate that the feature distributions of both the overall images and local defects generated by the Real-Adapter on the NEU-SEG and MT datasets closely align with those of the original images.

fect outperform diffusion-based models in generating more accurate and detailed features.

4.4.4. Downstream task augmentation experiments

To evaluate the contribution of generative models to downstream task augmentation, we conduct data augmentation experiments across four datasets to assess their impact on the performance of industrial inspection tasks. These experiments employ DeepLabv3+ as the detection model. Each training dataset comprises 600 images, including 300 original images and 300 generated images. The final experimental results are presented in Table 6. To ensure the robustness of these findings against random initialization and optimization variances, all experiments were evaluated over 5 independent runs with different random seeds. Statistical analysis via paired t-tests confirms that Real-Adapter achieves stable and significant improvements in detection performance across all four datasets at a 95% confidence level ($p < .05$). Specifically, in terms of mAP, the framework outperforms the second-ranked model by 5.083, 6.715, 3.584, and 4.554, respectively. These results demonstrate that the data generated by Real-Adapter are highly similar to real industrial data and provide an effective data augmentation strategy. As shown in Fig. 12, DeepLabv3+ augmented with Real-Adapter exhibits the most

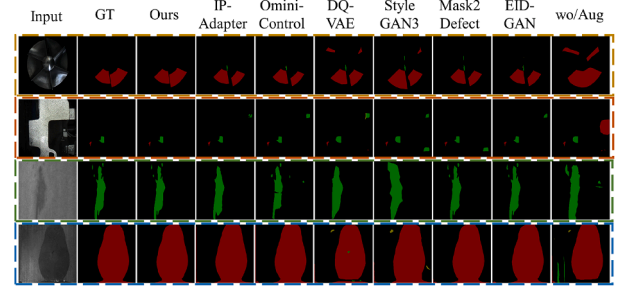


Fig. 12. The visualization of segmentation results from downstream detection models augmented with various generative models. The gray, orange, green, and blue dashed line refers to the segmentation masks from ID, CBD, NEU-SEG, and MT datasets.

accurate segmentation of the defect mask, thereby demonstrating its effectiveness in enhancing downstream tasks.

Since generated images inevitably contain certain levels of noise, excessive integration of synthetic data into the training dataset can diminish the augmentation efficacy by introducing significant artifacts. Therefore, we evaluated varying mixing proportions of generated data during training. As illustrated in Fig. 13, conventional generative methods (e.g., EID-GAN, Mask2Defect) typically exhibit a performance peak around a 1:1 real-to-generated ratio. Beyond this threshold (e.g., at 4:5), their augmentation performance distinctly deteriorates. This decline suggests that an overabundance of synthetic data from these methods introduces distribution shifts and accumulated artifacts, which ultimately overwhelm the discriminative features of real samples.

In contrast, Real-Adapter achieves steady performance gains even at a 2:3 generated data ratio. This resilience indicates that our architecture effectively bridges the domain gap by producing high-fidelity, low-noise, and domain-consistent samples. While traditional methods strictly limit augmentation ratios to prevent performance degradation, Real-Adapter allows for greater flexibility. Consequently, we recommend adopting higher mixing ratios (e.g., $> 1:1$) in severely data-scarce industrial scenarios to maximize downstream detection performance while avoiding the performance collapse typical of conventional generative augmentation.

In real-world industrial inspection tasks, accurate defect localization is essential for subsequent robotic handling and quality evaluation. To verify the impact of the Real-Adapter framework on downstream defect localization, we utilize YOLOv11-seg to assess augmentation effectiveness for simultaneous localization and instance segmentation. As shown in Table 7, training YOLOv11-seg with data synthesized by Real-Adapter

Table 6

The augmentation comparison of seven leading methods on DeepLabv3+ for defect segmentation.

Methods	ID		CBD		NEU-SEG		MT	
	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)
wo/Aug	58.941±0.924	51.765±0.815	52.572±1.021	41.678±0.854	61.713±0.963	50.866±0.772	47.383±1.105	39.682±0.912
EID-GAN	76.833±0.742	67.154±0.685	69.192±0.812	55.877±0.733	<u>80.144±0.618</u>	<u>70.292±0.554</u>	<u>65.983±0.792</u>	<u>54.714±0.648</u>
Mask2Defect	75.052±0.684	67.467±0.627	71.263±0.598	64.307±0.562	<u>79.185±0.672</u>	<u>71.232±0.591</u>	<u>65.259±0.735</u>	<u>55.297±0.612</u>
StyleGAN3	68.566±1.152	60.593±1.034	60.482±1.215	51.545±1.088	77.192±0.985	67.267±0.924	62.112±1.246	51.714±1.103
DQ-VAE	75.105±0.865	67.080±0.792	64.921±0.954	51.654±0.866	78.233±0.841	68.476±0.785	60.687±0.982	50.963±0.854
OminiControl	79.538±0.521	70.416±0.485	70.531±0.682	61.973±0.614	72.245±0.756	61.933±0.712	56.506±0.884	45.054±0.825
IP-Adapter	<u>78.064±0.584</u>	<u>69.936±0.532</u>	<u>71.599±0.627</u>	<u>62.148±0.583</u>	74.483±0.724	63.372±0.685	57.629±0.851	47.820±0.796
Ours	84.621±0.315	75.213±0.284	78.314±0.362	69.215±0.318	83.728±0.385	75.494±0.342	70.537±0.415	61.422±0.376

Note: Bold, Italic, and Underline symbolize the top-three performance of each metric. Results are reported as Mean ± Standard Deviation evaluated over 5 independent runs with different random seeds. Paired t-tests indicate that our Real-Adapter achieves statistically significant improvements over the best-performing baselines at a 95% confidence level ($p < 0.05$).

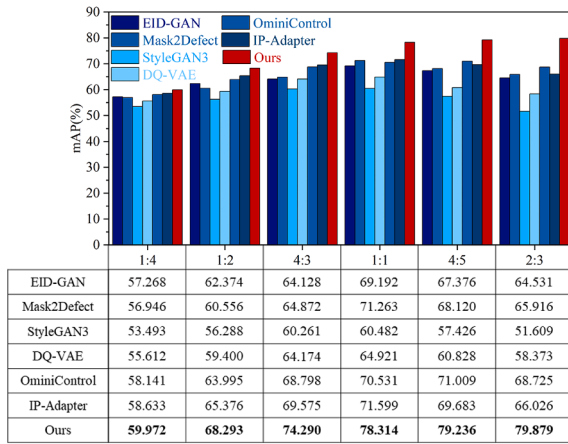


Fig. 13. Augmentation experiments with varying proportions of generated data in the training dataset. Experimental results demonstrate that the feature distribution generated by Real-Adapter closely aligns with the original data, enabling a higher real-to-generated ratio for data augmentation without altering the underlying distribution.

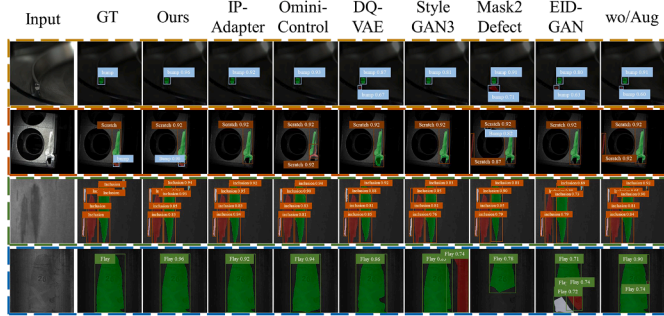


Fig. 14. The visualization of instance segmentation and localization results from YOLOv11-seg augmented with various generative methods. The brown, orange, green, and blue dashed lines correspond to the detection bounding boxes and masks from the ID, CBD, NEU-SEG, and MT datasets, respectively.

yields consistent quantitative improvements. To guarantee the reliability of these findings, all instance segmentation experiments were evaluated over 5 independent runs with different random seeds. For instance, on the ID dataset, our method achieves a mean mAP of 86.752% and a mean mIoU of 78.415%, outperforming the second-ranked method (OminiControl) by 4.638% and 4.853%, respectively. Notably, diffusion-based baselines (e.g., OminiControl and IP-Adapter) exhibit competitive downstream performance on the structurally complex ID and CBD datasets. Conversely, specialized industrial data augmentation methods

(e.g., EID-GAN and Mask2Defect) present an advantage on the NEU-SEG and MT datasets, where backgrounds are relatively uniform. Despite these varying performance trends among baselines, Real-Adapter demonstrates statistically significant improvements ($p < .05$) across all evaluated datasets. Furthermore, as visualized in Fig. 14, YOLOv11-seg augmented by Real-Adapter produces tighter bounding boxes and more continuous masks for fine-grained defects. This indicates that the domain-specific features embedded by our decoupled adapters effectively enhance the geometric localization capabilities of downstream models.

4.4.5. Ablation experiments

To demonstrate the effectiveness of our proposed modules, we conduct ablation experiments on each module on the CBD dataset. In this experiment, we remove each module separately, and the experimental results and visualizations are presented in Table 8 and Fig. 15(a). The experiments reveal the impact of each module on model performance. Specifically, removing the SIA branch results in a substantial decline in the authenticity of the generated image, with FID increasing by 40.096 and IS decreasing by 1.210. The model's detection performance also declines substantially, with mIoU dropping by 8.726 and mAP by 8.789. As shown in Fig. 15(b), (c), the generated image exhibits noticeable distortions when the SIA branch is removed. This is due to SIA providing a large number of domain-specific scene features. In addition, omitting local or location information in the LLIA branch leads to a slight decrease in the overall authenticity of the generated image. However, the effectiveness of augmentation decreases considerably, suggesting that this information is critical for downstream task augmentation. Furthermore, as shown in Fig. 15, when local features are missing, defect forms lack authenticity, and when location information is absent, defects emerge in implausible locations.

Furthermore, we attempt to optimize the model without utilizing the TA loss. Although the FID does not exhibit a substantial change, the performance benefits for downstream tasks decline notably, with the mAP and mIoU metrics experiencing decreases of 11.059 and 11.040, respectively. This result demonstrates that the TA loss provides a more potent optimization for defect feature authenticity. Finally, we evaluate the effectiveness of different TA-loss supervision variants. Replacing the noise-aware pseudo-labels with clean-image masks leads to a slight but consistent drop in both generative quality and augmentation performance. Specifically, compared with the noisy-mask variant, the clean-mask variant increases FID from 42.597 to 48.235, while decreasing IS from 8.152 to 7.829, mAP from 78.314 to 75.748, and mIoU from 69.215 to 67.543. At the same time, both TA-loss variants still outperform the w/o-TA setting, confirming the overall effectiveness of task-aware supervision. This can be attributed to the fact that clean-image masks are not fully condition-matched to the corrupted states encountered during diffusion training. Even within the selected early noisy steps, part of the fine defect geometry may become weakened

Table 7

The augmentation comparison of seven leading methods on YOLOv11-seg for instance segmentation and localization.

Methods	ID		CBD		NEU-SEG		MT	
	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)	mAP(↑)	mIoU(↑)
wo/Aug	62.145±1.052	54.321±0.984	56.812±1.115	45.912±1.085	65.412±0.985	54.215±0.912	50.814±1.205	42.145±1.154
EID-GAN	77.512±0.821	69.102±0.765	70.145±0.785	59.814±0.814	82.145±0.658	73.814±0.612	69.512±0.795	59.124±0.765
Mask2Defect	76.841±0.792	68.125±0.812	71.512±0.752	62.145±0.763	83.512±0.615	75.124±0.584	67.814±0.812	57.415±0.785
StyleGAN3	71.425±1.124	63.814±1.056	63.512±1.254	54.124±1.185	79.512±1.025	69.814±0.965	65.124±1.156	54.812±1.098
DQ-VAE	78.214±0.875	69.512±0.798	68.214±0.895	57.312±0.842	80.124±0.854	70.415±0.812	62.415±0.954	52.145±0.915
OminiControl	82.114±0.612	73.562±0.584	73.814±0.654	64.912±0.621	75.214±0.745	64.125±0.715	58.415±0.845	48.512±0.812
IP-Adapter	80.453±0.645	71.824±0.615	75.612±0.632	66.425±0.598	77.412±0.712	66.512±0.685	60.124±0.815	50.412±0.795
Ours	86.752±0.325	78.415±0.298	81.345±0.341	72.814±0.315	86.124±0.355	78.512±0.324	74.812±0.385	65.214±0.356

Note: The colors brown, green, and blue symbolize the top-three performance of each metric. Results are reported as Mean ± Standard Deviation evaluated over 5 independent runs with different random seeds. Paired t-tests indicate that our Real-Adapter achieves statistically significant improvements over the best-performing baselines at a 95% confidence level ($p < .05$).

Table 8

The ablation experiments that demonstrate the effectiveness of each proposed module.

	SIA	LLIA w/ H_{lc}	LLIA w/ H_t	TA loss noised GT	TA loss clean GT	Generative		Augmentation	
						FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)
×	✓	✓	✓	×	×	82.693	6.942	69.588	60.426
✓	×	✓	✓	✓	×	61.350	7.726	64.499	54.381
✓	✓	×	✓	✓	×	50.268	7.543	66.127	57.642
✓	✓	✓	×	×	×	52.987	7.332	67.255	58.175
✓	✓	✓	✓	×	✓	48.235	7.829	75.748	67.543
✓	✓	✓	✓	✓	×	42.597	8.152	78.314	69.215

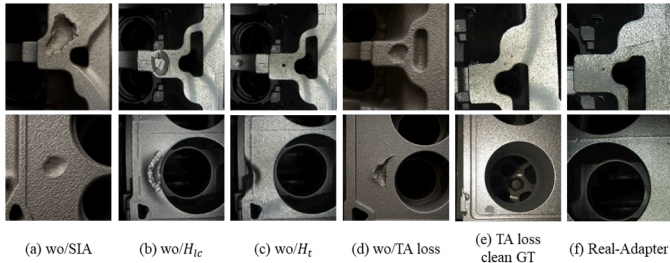


Fig. 15. The visualization of generative results of Real-Adapter after removing the different modules. The visual evidence indicates that omitting the SIA, LLIA, or TA loss leads to images that respectively display issues like artificial textures, inauthentic defects, and inaccurate defect placement. These results demonstrate that each module is integral to the Real-Adapter framework.

or partially ambiguous. Enforcing a complete clean mask under such conditions may therefore introduce a mild supervision mismatch and slightly biased gradients. In contrast, pseudo-labels generated from the corresponding noisy inputs are more consistent with the defect evidence actually preserved at each diffusion step, yielding a better-aligned task-aware signal. As a result, the noisy-mask variant performs slightly better than the clean-mask variant. The performance gap remains limited because the selected early noisy steps still preserve most of the defect topology, so clean-mask supervision is still largely valid.

To explicitly isolate the contribution of each conditioning source, we conducted additive ablation experiments (Fig. 16 and Table 9). Mechanistically, relying solely on textual location prompts yields the poorest generative quality (FID: 112.340), as the model hallucinates incorrect global textures and defect styles. When only local defect information is provided, valid defect features emerge, but the global background is severely distorted due to the lack of domain-specific scenario knowledge. Conversely, isolating the SIA branch successfully reconstructs clear and realistic global industrial textures (FID improves to 75.432); however, it fails to synthesize specific defects entirely. Due to severe structural distortions or the complete absence of targeted defects, im-

Table 9

Quantitative additive ablation results isolating the effect of each conditioning source on the CBD dataset.

Method	SIA	Local	Location	FID (↓)	IS (↑)
Location Only			✓	112.340	5.980
Local Info Only		✓		98.765	6.450
SIA Only	✓			75.432	7.120
Real-Adapter	✓	✓	✓	42.597	8.152

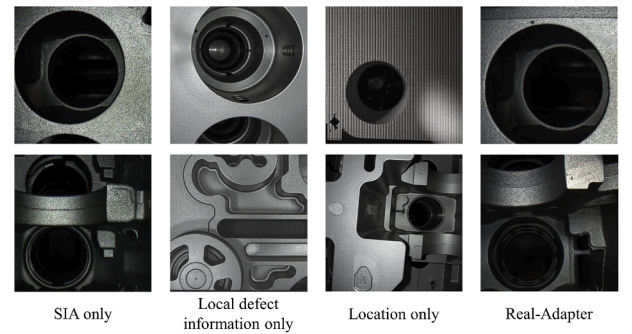


Fig. 16. Visual comparison of isolated conditioning sources. From left to right: SIA only (realistic global texture, but no specific defects), Local information only (valid defects, but distorted global texture), Location only (incorrect texture and defect styles), and Real-Adapter (high-fidelity background with accurate defects).

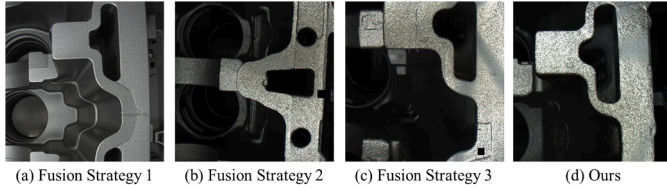
ages generated by these isolated sources are of insufficient quality to be utilized for downstream data augmentation. Only our complete Real-Adapter synergizes these priors effectively, balancing high-fidelity background reconstruction with precise defect generation.

To verify the necessity of our decoupled fusion between SIA (global scene semantics) and LLIA (local defect cues), we compare three coupled baselines under identical settings (same backbone, splits, seed, and hyperparameters). Fusion Strategy 1 injects both SIA and LLIA into the encoder and decoder (Eq. (9)); Strategy 2 injects SIA into the encoder

Table 10

Ablation on control signal fusion strategies for SIA and LLIA branches.

Methods	FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)
Fusion Strategy 1	79.832	7.927	62.904	54.225
Fusion Strategy 2	53.125	8.043	71.482	60.259
Fusion Strategy 3	57.623	8.107	72.518	62.341
Ours	42.597	8.152	78.314	69.215

**Fig. 17.** The visualization of generative results of Real-Adapter using four various control signal fusion strategies. This visual representation demonstrates that the images generated via our fusion strategy exhibit superior realism regarding texture and fine details.

and couples SIA + LLIA by summation in the decoder (Eq. (10)); Strategy 3 couples SIA and LLIA by concatenation in the decoder (Eq. (11)). As shown in Table 10, our method achieves the best overall performance (FID 42.597, IS 8.152, mAP 78.314, mIoU 69.215). Compared with Strategy 1, our design improves FID by 37.235 and yields large gains on downstream metrics (mAP +15.410, mIoU +14.990), suggesting that injecting LLIA at encoder of the denoising U-Net can distort global structure formation and introduce background artifacts that harm both fidelity and augmentation utility. Against the best coupled decoder fusion (Strategy 2 in FID), we further reduce FID from 53.125 to 42.597 (−10.528), while improving mAP from 71.482 to 78.314 (+6.832) and mIoU from 60.259 to 69.215 (+8.956). Similarly, relative to Strategy 3 (concat), our method improves mAP by 5.796 and mIoU by 6.874, indicating better preservation of fine-grained defect cues. These gains are consistent with our mechanism analysis: coupled fusion under a shared softmax normalization tends to allocate most attention mass to dense scene tokens, suppressing sparse defect tokens and attenuating defect-related gradients, which manifests as defect dilution and location drift. Fig. 17 further provides intuitive evidence for the above findings. In Fig. 17(a) (Fusion Strategy 1), the overall surface texture deviates noticeably from real metallic materials, exhibiting less realistic reflectance and unnatural patterns. In Fig. 17(b) and (c) (Fusion Strategy 2 and 3), the synthesized defect morphology or location becomes less faithful and shows a clear drift from the intended region. By contrast, Fig. 17(d) (Ours) produces a more natural global style with realistic metallic texture, while simultaneously preserving defect morphology and precise placement, demonstrating the advantage of our decoupled global–local fusion for industrial defect generation.

To investigate the optimal parameter configuration and the relationship between model architecture and performance, we conduct ablation studies on four critical hyperparameters, validated on the CBD dataset. Table 11 illustrates the impact of the depths of the SIA (d_{SIA}) and LLIA (d_{LLIA}) branches on model performance. As pivotal architectural parameters, these depths govern the efficacy of domain-specific knowledge extraction within the generative model. Increasing the depth enhances the model's theoretical capacity for feature extraction but also expands the number of trainable parameters. This expansion can hinder effective training when only a limited number of industrial samples are available. Consequently, our objective is to identify an optimal trade-off between information extraction quality and training efficiency. Experimental results indicate that a SIA depth of 4 and an LLIA depth of 3 yield optimal performance in both image generation and downstream task enhancement. For instance, increasing the SIA depth to 5 introduces 17.042M additional parameters, imposing a significant training burden and re-

sulting in performance degradation: a 17.681 increase in FID, alongside decreases of 0.043 in IS, 4.743 in mAP, and 5.997 in mIoU. Conversely, reducing the SIA depth to 3 decreases parameters but leads to suboptimal performance due to insufficient information extraction capacity. A similar trend is observed for the LLIA depth, confirming the necessity of balancing architectural complexity with the constraints of limited data.

Table 12 presents an analysis of the impact of β , which balances the denoising loss (L_θ) and the TA loss (L_{TA}), and its decay coefficient d_β on model performance. The coefficient d_β defines the amount by which β is reduced every 1×10^4 training iterations. During the early training stages, a higher weight on the denoising loss is necessary to ensure the model focuses on global feature generation and overall image structure. In later stages, increasing the proportion of the TA loss enhances the model's capacity for generating detailed defects. As shown in the table, the optimal performance is achieved with $\beta = 0.9$ and $d_\beta = 0.05$, followed by the configuration $\beta = 0.9, d_\beta = 0.1$. Compared to suboptimal settings where the denoising loss weight is too small or decays too rapidly (e.g., $\beta = 0.6, d_\beta = 0.1$), the optimal parameters yield a substantial improvement: a reduction in FID by 46.815, and increases in IS, mAP, and mIoU of 1.838, 16.930, and 17.592, respectively. These results underscore the importance of prioritizing global generation in early training iteration and gradually incorporating TA loss to refine detail generation in later stages.

Table 13 presents the effect of varying the weighting factor α , which balances the L_{dice} and $L_{boundary}$ components in our TA loss, on the model's performance. The results demonstrate that the model achieves optimal performance at $\alpha = 0.6$. With this configuration, the FID score is 1.855 points lower than the next best setting, while the IS, mAP, and mIoU metrics are increased by 0.021, 1.593, and 0.678, respectively. Nevertheless, the overall performance exhibits considerable robustness to changes in α , indicating that the weighting of these loss components has a limited influence on the final model efficacy.

4.5. Discussion

This paper proposes Real-Adapter, a novel framework for integrating diffusion-based large generative foundation models into industrial data augmentation. Unlike classical approaches focused on optimizing GAN- or VAE-based architectures, our work attempts to integrate diffusion models into industrial data augmentation, establishing a novel paradigm that opens new research avenues in this field. Overall, Real-Adapter addresses two key challenges in adapting diffusion-based large generative models to industrial image generation and augmentation: domain-specific knowledge gaps and insufficient fine-grained detail synthesis.

We conduct experiments using two self-constructed datasets (ID and CBD) and two publicly available benchmarks (NEU-SEG and MT). Real-Adapter successfully generates synthetic data that faithfully replicates the characteristics of real-world industrial inspection scenarios. Notably, on the ID and CBD datasets, Real-Adapter demonstrates robust performance in interpreting and synthesizing intricate domain-specific industrial environments. As shown in Table 3, Real-Adapter achieves FID scores surpassing the second-best baseline by impressive margins of 16.834 and 14.645 on the ID and CBD datasets, respectively. Fig. 5 presents a comparison of the generation results from Real-Adapter and several representative generative paradigms. The images synthesized by Real-Adapter demonstrate a strong adherence to physical principles in both their overall structure and detailed defect morphology. Furthermore, Fig. 6 visualizes the feature information extracted by the SIA and LLIA branches, illustrating their respective capabilities in capturing professional and detailed informational features. As illustrated in Fig. 7, the spatial representations of both the overall structure and fine-grained defects in images generated by Real-Adapter closely align with those of real-world samples. Fig. 8 demonstrates that Real-Adapter achieves superior generation performance under extreme data scarcity conditions compared to other representative generative methods. Fur-

Table 11The sensitivity analysis of SIA depth (d_{SIA}) and LLIA depth (d_{LLIA}).

SIA (LLIA depth = 3)						LLIA (SIA depth = 4)					
d_{SIA}	FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)	Param (M)	d_{LLIA}	FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)	Param (M)
1	82.856	7.242	69.351	60.287	0.080	1	50.863	7.595	67.153	57.284	0.031
2	72.540	7.886	72.582	61.258	0.376	2	47.237	7.992	72.325	61.615	0.146
3	61.581	7.992	72.215	64.801	1.491	3	42.597	8.152	78.314	69.215	0.589
4	42.597	8.152	78.314	69.215	5.818	4	42.994	8.129	78.302	69.301	2.326
5	60.278	8.109	73.571	63.218	22.860	5	44.626	8.153	74.328	65.107	9.209
6	100.919	6.518	50.123	41.952	90.499	6	47.627	8.023	66.259	57.203	36.606

Table 12Sensitivity analysis of weight balance β and weight decay d_β on generative and downstream augmentation performance.

β	d_β	FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)
0.9	0.1	47.578	8.025	76.892	67.137
0.8	0.1	58.543	7.528	71.027	62.457
0.7	0.1	79.213	6.892	68.772	59.214
0.6	0.1	89.412	6.314	61.384	51.623
0.9	0.05	42.597	8.152	78.314	69.215
0.8	0.05	46.209	7.951	76.429	66.462
0.7	0.05	56.275	7.423	69.285	61.723
0.6	0.05	69.674	7.025	69.334	60.281

Table 13Sensitivity analysis of TA loss weight α on generative and downstream augmentation performance.

α	FID(↓)	IS(↑)	mAP(↑)	mIoU(↑)
0.9	51.124	7.624	73.208	65.651
0.8	47.764	7.959	73.268	66.237
0.7	44.452	8.131	76.721	68.537
0.6	42.597	8.152	78.314	69.215
0.5	43.578	8.012	76.582	68.237
0.4	44.164	8.005	76.381	68.171
0.3	49.578	7.821	74.014	66.137
0.2	54.578	7.995	72.691	64.137

thermore, Fig. 9 highlights Real-Adapter's exceptional performance in synthesizing diverse defect types, underscoring its adaptability to varying defect morphological variations.

To evaluate the generalization capability of Real-Adapter, we conduct generalization experiments on the NEU-SEG and MT datasets. Real-Adapter outperforms the second-best model, achieving improvements in FID scores by 1.112 and 3.531 and in IS by 0.602 and 0.277, respectively, demonstrating robust generalization performance. In downstream task augmentation experiments, we enrich the four datasets with synthetic data. The enhanced detection models achieve state-of-the-art performance, with mAP improvements of 5.083, 6.715, 3.584, and 4.554 over the second-best approach across the four datasets. As shown in Fig. 13, the synthetic data generated by Real-Adapter exhibits considerably lower noise levels compared to baseline methods. Additionally, ablation studies confirm the necessity of each module. Removing the SIA branch results in substantial degradations: FID and IS metrics decline by 40.096 and 1.210, while mIoU and mAP decrease by 8.726 and 8.789, respectively. This underscores the critical role of domain-specific industrial knowledge encoded by the SIA branch in ensuring realistic image synthesis and enhancing downstream detection accuracy. Similarly, removing the LLIA branch and TA loss leads to a sharp decline in downstream task performance, validating their importance in refining fine-grained defect realism and optimizing data augmentation efficacy. Finally, Tables 11–13 present the ablation studies on key hyperparameters. We investigate the influence of these hyperparameters on model

performance and elucidate the underlying mechanisms, thereby providing clear guidance for parameter tuning.

However, our work has certain limitations. In real-world industrial inspection scenarios, variations in environmental and sensor parameters, such as camera configurations, lighting intensity, and material reflectivity, critically influence both data acquisition and downstream inspection performance. The current version of Real-Adapter cannot dynamically adapt to replicate data across diverse operational conditions. In addition, industrial defects sometimes encompass logical defects, such as missing components or incorrect assembly configurations. Currently, synthesizing purely logical defects remains a challenge that requires broader structural reasoning extending beyond surface anomalies. Future research will prioritize extending the framework to simulate data generation under varying illumination conditions, camera parameters, and other domain-specific variables, thereby enhancing its practical applicability. Furthermore, extending the current architecture to explicitly comprehend and generate complex logical industrial defects constitutes another important direction for our future work.

5. Conclusion

This paper introduces a lightweight generative adapter integrated with a large generative foundation model to synthesize industrial component defect data characterized by intricate textures. The proposed framework adopts a more robust diffusion-based generation method to replace the GAN-based generation method and substantially improves downstream detection performance while addressing two critical challenges in diffusion-based industrial data augmentation: domain-specific knowledge gaps between foundation models and real industrial scenarios, and insufficient fine-grained defect synthesis of the foundation model. To bridge the domain-specific knowledge gap, we propose an SIA branch that extracts industrial scenario-specific information. This information is aligned with the foundation model's latent space, enabling the generation of data that preserves realistic industrial structural and contextual attributes. To enhance the authenticity of synthesized fine-grained defects, we design an LLIA branch that captures defect geometric properties and spatial distribution patterns, facilitating precise defect synthesis at arbitrary locations. Both SIA and LLIA are implemented via streamlined feature extraction layers, forming a parameter-efficient architecture to circumvent the inefficiency of retraining foundation models under limited data constraints. During training, the foundation model remains frozen, while only the SIA and LLIA branches are exclusively optimized to align domain-specific features with the diffusion framework. Furthermore, a TA loss function is introduced to maximize the efficacy of data augmentation. Extensive experiments demonstrate the framework's superiority in synthetic data quality and downstream detection accuracy. However, the current framework exhibits limitations in dynamically adapting to significant variations in physical environments, such as fluctuating lighting intensity and diverse materials. Furthermore, Real-Adapter lacks the capability to generate logical defects that arise in industrial production. Consequently, future work prioritizes extending the framework to simulate and generalize across these variable manufacturing conditions and investigating methods that enable the diffusion model to comprehend spatial logic and generate

logical defects, thereby improving its practical applicability in complex industrial scenarios.

CRedit authorship contribution statement

Chenghan Pu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft; **Ziyu Lin:** Software, Validation, Investigation; **Kaijun Zhang:** Methodology, Software, Formal analysis, Writing – review & editing; **Xiang Wu:** Resources, Data curation, Investigation; **Zikuan Li:** Methodology, Formal analysis, Writing – review & editing; **Le Lu:** Investigation; **Jun Wang:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jun Wang reports financial support was provided by National Natural Science Foundation of China. Reports a relationship with that includes: Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.eswa.2026.132452](https://doi.org/10.1016/j.eswa.2026.132452)

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